

Weekly Notice 1

A few remarks:

1. *Literature:* We will mainly use the lecture notes *Stochastic Processes* (Jan Pedersen) which are available in pdf format on the course home page. Additionally, you can buy a hard copy in the book store in the basement of the math department. **Make sure you get the 2026 version.**
2. I ($\equiv$ Jan) will act as both instructor (forelæser) and TA (instruktør). My office is B4.26 (aka 1535-426).
3. There are two mandatory assignments in the course. The first one appears below.
4. For information about the lectures see the folder *The Lectures* on the course home page.

Exercises, first week: ("Exercise x" refers to Exercise x in the lecture notes.)

1. Exercise 1. Note that you don't have to prove anything.
2. Exercise 28(1)–(2).
3. Exercise 29(1).
4. Exercise 2. Question (2) relies on Appendix F.
5. Exercise 4. Question (3) relies on Appendix F.

The first mandatory assignment consists of the following four exercises.

1. Consider i.i.d. real random variables $Y_1, Y_2, \dots$ with common density f given by

$$f(y) = \begin{cases} e^{-2y}, & y \geq 0, \\ \frac{1}{2}e^y & y < 0. \end{cases}$$

Let $S_0 = 0$ and define $(S_n)_{n \geq 1}$ as

$$S_n = Y_1 + \dots + Y_n, \quad \text{for } n \in \mathbb{N}.$$

Also define

$$\mathcal{F}_n = \begin{cases} \{\emptyset, \Omega\} & \text{for } n = 0 \\ \sigma(Y_1, \dots, Y_n) & \text{for } n \in \mathbb{N}. \end{cases}$$

- (a) For $n \in \mathbb{N}_0$ determine $\mathbb{E}[e^{\alpha S_{n+1}} | \mathcal{F}_n]$ for all $\alpha \in \mathbb{R}$.
- (b) Determine $\mathbb{P}(S_n < 0 < S_{n+1} | \mathcal{F}_n)$ for $n \in \mathbb{N}$.

2. Let the state space be $S = \mathbb{N}_0 = \{0, 1, \dots\}$ and for $i \in S$ let $p_i \in [0, 1]$. Consider the Markov chain with state space S specified as follows: A particle which in state $i \in S$ moves to state $i + 1$ in one step with probability p_i and to state 0 with probability $1 - p_i$.
 - (a) Argue that the chain is irreducible if $p_i \in (0, 1)$ for all $i \in S$. Also, indicate the transition diagram in this case.
 - (b) Determine p_{02}^4 . For $n \in \mathbb{N}$ determine $\mathbb{P}_0(T_0 \geq n)$ and $\mathbb{P}_0(T_0 = n)$.
 - (c) Show that state 0 is recurrent if and only if $\prod_{i=0}^{\infty} p_i = 0$.
 - (d) Give a concrete example of probabilities $(p_i)_{i=0}^{\infty}$ such that $p_i \in (0, 1)$ for all $i \in S$ and the chain is transient.
 - (e) Assume that $p_i \in (0, 1)$ for all $i \in S$. Show that there exists a unique stationary distribution if and only if $\sum_{n=0}^{\infty} (p_0 p_1 \cdots p_n) < \infty$. Determine the stationary distribution when it exists.

3. Consider Proposition 6.4 in the lecture notes.

(a) Prove (6.4).

Let $i \in S$ and μ denote a probability measure on S . Assume $\mathbb{P}_\mu(T_i < \infty) = 1$ and $\mathbb{P}_i(T_i < \infty) = 1$.

(b) Show that T_i^n is finite $\mathbb{P}_\mu$ -a.s. for every $n \in \mathbb{N}$.

(c) Show that T_i^1 and $T_i^2 - T_i^1$ are independent under $\mathbb{P}_\mu$ and that the distribution of $T_i^2 - T_i^1$ under $\mathbb{P}_\mu$ is the same as the distribution of T_i^1 under $\mathbb{P}_i$.

Essentially, the above proves Proposition 6.4. The only remaining step is to extend (c) above by induction to get Proposition 6.4(2)(b)–(c).

4. Let $S = \{1, 2, 3, 4, 5, 6\}$ and

$$P = \begin{pmatrix} 0.1 & 0.4 & 0.1 & 0 & 0 & 0.4 \\ 0 & 0.2 & 0.3 & 0.5 & 0 & 0 \\ 0 & 0 & 0.2 & 0.3 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0.2 & 0.2 & 0.1 \\ 0 & 0 & 0.8 & 0 & 0.2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

- (a) Argue that there exists a unique stationary distribution, determine it, and determine $\mathbb{E}_i[T_i]$ for all $i \in S$.
- (b) Let $\tau = \inf\{n \geq 2: X_n = 5\}$. Argue that $\mathbb{P}_1(\tau < \infty) = 1$ and determine $\mathbb{E}_1[X_{\tau+1}]$.